

Distributed Optimization for Machine Learning

Lecture 16 - Decentralized SGD and its convergence

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SGD sampling: Theory vs. practice

Theory: With replacement

- Each iteration draws an independent sample ξ^t uniformly from the dataset.

$$\xi^t \sim \text{Uniform}\{\xi_1, \dots, \xi_n\}$$

- The stochastic gradient is $g(\mathbf{x}^t; \xi^t)$, and the update rule:

$$\mathbf{x}^{t+1} = \mathbf{x}^t - \eta_t g(\mathbf{x}^t; \xi^t)$$

- Every iteration is independent - the same sample may appear multiple times in an epoch.

Practice: Without replacement

- Modern libraries (PyTorch) **shuffle the dataset once per epoch**. Each sample is used exactly once in an epoch:

$$\{\xi_1^{(e)}, \dots, \xi_n^{(e)}\} = \text{Shuffle}(\{\xi_1, \dots, \xi_n\})$$

- Iterates follow the order:

$$\mathbf{x}^{t+1} = \mathbf{x}^t - \eta_t g(\mathbf{x}^t; \xi_i^{(e)})$$

- The samples are correlated within an epoch but nearly independent across epochs.



Why this difference exists (and why it's okay)

■ Theoretical reason:

- Sampling with replacement makes $\{g(\mathbf{x}^t; \xi^t)\}$ i.i.d.
- Enables clean proofs for convergence and variance bounds.

$$\mathbb{E}[g(\mathbf{x}^t; \xi^t)] = \nabla F(\mathbf{x}^t)$$

■ Practical reason:

- Sampling without replacement ensures all data are used each epoch.
- Reduces redundancy and improves practical convergence speed.
- For large datasets, dependence within an epoch is negligible.

■ Empirical observation:

- "Without-replacement" SGD often converges faster per epoch.
- "With-replacement" SGD remains the default for theoretical analysis.

Takeaway: Sampling with replacement $\Rightarrow$ i.i.d. gradients, easier theory. Without replacement $\Rightarrow$ efficient practice, same limit behavior.



Coding snapshot: Sampling protocols in SGD

Sampling with vs. without replacement

```
# Batch size
batch_size = 32

# ---- Sampling WITH replacement (theoretical SGD) ----
idx = np.random.choice(len(data), size=batch_size, replace=True)
batch_X = data[idx]

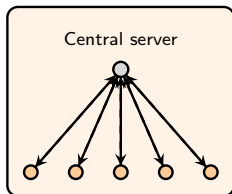
# ---- Sampling WITHOUT replacement (practical SGD) ----
idx = np.random.choice(len(data), size=batch_size, replace=False)
batch_X = data[idx]
```

- `replace=True` $\Rightarrow$ i.i.d. sampling (matches theory).
- `replace=False` $\Rightarrow$ epoch-wise shuffling (common in PyTorch).



Review of Parallel SGD: synchronize every time

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}), \quad \text{where } f_i(\mathbf{x}) = \mathbb{E}_{\xi_i \sim D_i} [F(\mathbf{x}; \xi_i)].$$



Local data on nodes

$$\begin{aligned} g_i^k &= \nabla F(\mathbf{x}^k; \xi_i^k) && \text{(Local compt.)} \\ \mathbf{x}^{k+1} &= \mathbf{x}^k - \frac{\eta}{n} \sum_{i=1}^n g_i^k && \text{(Global comm.)} \end{aligned}$$

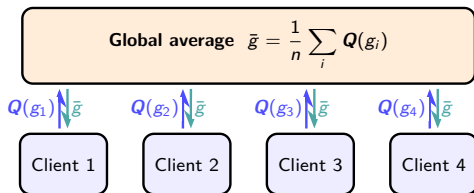
- All nodes synchronize (i.e., *globally average*) **every time**.
- Global average protocol: Ring-AllReduce or central server

Communication cost is $\mathcal{O}(n)$ which is high when n is large.



Last-lecture: Compressed SGD

Compressed SGD: Each node i communicates its compressed gradient.



$$\begin{aligned} g_i^k &= \nabla F(\mathbf{x}^k; \xi_i^k) && \text{(Local compt.)} \\ \mathbf{x}^{k+1} &= \mathbf{x}^k - \frac{\eta}{n} \sum_{i=1}^n Q(g_i^k) && \text{(Global comm.)} \end{aligned}$$

- Compressed methods: quantization, sparsification

Reduce the communication cost by reducing the dimensionality.

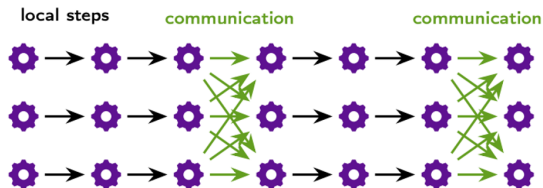


Last-lecture: Local SGD

Local SGD: Each node i performs τ -step SGD before averaging.

$$\mathbf{x}_i^{(s+1)} = \mathbf{x}_i^{(s)} - \eta \nabla F(\mathbf{x}_i^{(s)}; \xi_i^{(s)}), \quad s = 1, \dots, \tau \quad (\text{Local updates})$$

$$\mathbf{x}^{k+1} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i^{(\tau)} \quad (\text{Global comm.})$$



Reduce the temporal communication cost by τ .



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Decentralized SGD: reducing spatial communication

Convergence of decentralized SGD

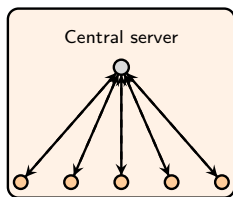


From local SGD to decentralized SGD: Why?

- **Local SGD:** Each worker performs several local updates before synchronizing with a **central server**. This reduces **temporal communication** but still requires global synchronization.
- **Limitation:** Every synchronization still needs central aggregation:
 $\mathcal{O}(n)$ **communication and vulnerable** to a single point of failure.
- **New question:** Can we **remove the central server** and achieve consensus only through **neighbor-to-neighbor communication**?
- **Goal:** Reduce **spatial communication cost** and make learning **more scalable, robust, and private**.

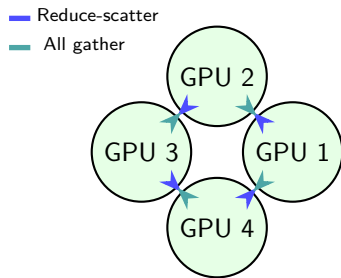


Global averaging protocols are expensive



Local data on nodes

Server: $\mathcal{O}(n)$ communication per round.



Ring-AllReduce: $\mathcal{O}(n)$ latency.

Observation: Both global protocols require $\mathcal{O}(n)$ complexity.



Moving beyond global averaging

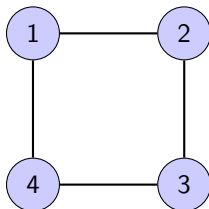


Can we leverage **Gossip** to reduce the spatial communication cost?



Neighbor set

- **Setup:** A network of n nodes connected by a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.



- **Node set $\mathcal{V}$:**

$$\mathcal{V} = \{1, 2, 3, 4\}$$

- **Edge set $\mathcal{E}$:**

$$\mathcal{E} = \{(1, 2), (1, 4), (2, 3), (3, 4)\}$$

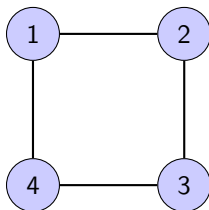
- **Neighbor set $\mathcal{N}_i$ for node i :**

$$\mathcal{N}_i = i \cup \{j : (i, j) \in \mathcal{E}\}$$



Examples of neighbor set

- **Setup:** A network of n nodes connected by a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.



- **Neighbor set $\mathcal{N}_1$ for node 1:**

$$\mathcal{N}_1 = \{2, 4\}$$

- **Neighbor set $\mathcal{N}_2$ for node 2:**

$$\mathcal{N}_2 = \{1, 3\}$$

- **Neighbor set $\mathcal{N}_3$ for node 3:**

$$\mathcal{N}_3 = \{2, 4\}$$

- **Neighbor set $\mathcal{N}_4$ for node 4:**

$$\mathcal{N}_4 = \{1, 3\}$$



Key idea of decentralized SGD



Instead of global averaging, partial averaging (gossip) with neighbors



Example of decentralized SGD on the ring graph

- Consider the 4-node ring graph given above.

Decentralized SGD on the 4-node ring graph:

1. At time k , each node i runs one (can be multiple) local SGD step:

$$\mathbf{x}_i^{(k+\frac{1}{2})} = \mathbf{x}_i^{(k)} - \eta \nabla F(\mathbf{x}_i^{(k)}; \xi_i^{(k)})$$

2. Every node i sent its current state $\mathbf{x}_i^{(k+\frac{1}{2})}$ to its neighbors $j \in \mathcal{N}_i$.
3. Every node i update its state by partial averaging with its neighbors:

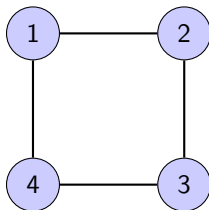
$$\text{For } \{l, m, i\} \in \mathcal{N}_i : \quad \mathbf{x}_i^{(k+1)} = \frac{1}{3}\mathbf{x}_i^{(k+\frac{1}{2})} + \frac{1}{3}\mathbf{x}_l^{(k+\frac{1}{2})} + \frac{1}{3}\mathbf{x}_m^{(k+\frac{1}{2})}$$

- Reduce the per-iteration comm. cost of PSGD from $\mathcal{O}(n)$ to $\mathcal{O}(1)$.



Review of weight matrix for the ring graph

- The weight matrix for the 4-node ring graph is static



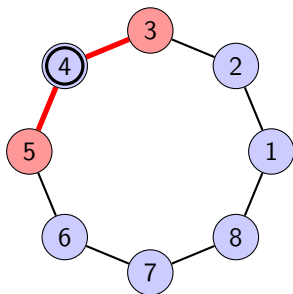
$$\mathbf{W} = \begin{bmatrix} 1/3 & 1/3 & 0 & 1/3 \\ 1/3 & 1/3 & 1/3 & 0 \\ 0 & 1/3 & 1/3 & 1/3 \\ 1/3 & 0 & 1/3 & 1/3 \end{bmatrix}$$

- $\mathbf{W}$ is **doubly stochastic** ($\mathbf{1}^T \mathbf{W} = \mathbf{1}^T$ and $\mathbf{W} \mathbf{1} = \mathbf{1}$).
- (Deterministic) Gossip: **reach consensus**
- Adding more edges to the graph can improve the consensus rate

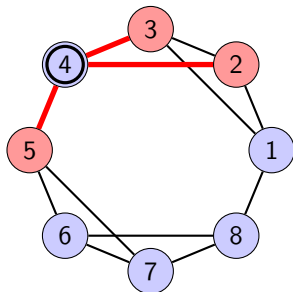


More graph topologies

- Decentralized SGD can be defined for all undirected graphs.



Ring graph:
every node has 2 neighbors



Every node has 3 neighbors.



Decentralized SGD: partial averaging via gossip

- DSGD: local SGD + partial averaging with neighbors

Decentralized SGD (DSGD)

Given a communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, at each round k ,

1. At time k , each node i runs one (can be multiple) local SGD steps:

$$\mathbf{x}_i^{(k+\frac{1}{2})} = \mathbf{x}_i^{(k)} - \eta \nabla F(\mathbf{x}_i^{(k)}; \xi_i^{(k)})$$

2. Every node i sent its current state $\mathbf{x}_i^{(k+\frac{1}{2})}$ to its neighbors $j \in \mathcal{N}_i$.
3. All nodes update their states by partial averaging with neighbors

$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}}$$

- **DSGD** reduces the spatial comm. cost when the graph is sparse.



DSGD has lower per-iter comm. cost than PSGD

- Experiment on a **256-GPU** cluster. Use Ring-AllReduce as global averaging protocol for PSGD.

Model	Ring-AllReduce (ms)	Partial Averaging (ms)
ResNet-50 (25.5M)	278	150

Table: Per-iteration *communication* runtime (lower is better [1]).

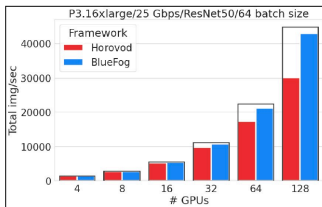
- **46% reduction** vs. PSGD.

[1] Y. Chen, K. Yuan, Y. Zhang, P. Pan, Y. Xu, and W. Yin, "Accelerating Gossip SGD with Periodic Global Averaging", ICML 2021.

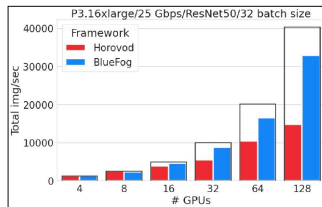


DSGD is more communication-efficient than PSGD

- DSGD (BlueFog) has **better scalability** than PSGD (Horovod) due to its small comm. overhead.



Small comm.-to-compt. ratio



Large comm.-to-compt. ratio



[2] B. Ying, K. Yuan, H. Hu, Y. Chen and W. Yin, BlueFog: Make decentralized algorithms practical for optimization and deep learning, arXiv: 2111. 04287, 2021

Summary of DSGD

Key idea: Instead of global averaging, partial averaging with neighbors

DSGD

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^k; \xi_i^k)$$

$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}}$$

- Each node i samples mini-batch ξ_i^k and computes $\nabla F(\mathbf{x}_i^k; \xi_i^k)$.
- Nodes synchronize with its neighbors.

Benefit: Reduce comm. cost to $O(d_{\max})$ where $d_{\max} = \max_i \{|\mathcal{N}_i|\}$.

How about its convergence?



Quadratic objectives: setup and average dynamics

Consider local quadratic objectives

$$f_i(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A}_i \mathbf{x} - \mathbf{b}_i^\top \mathbf{x},$$

and the decentralized update (no noise):

$$\mathbf{x}_i^{k+1} = \sum_{j=1}^n w_{ij} \left(\mathbf{x}_j^k - \eta (\mathbf{A}_j \mathbf{x}_j^k - \mathbf{b}_j) \right), \quad \text{where } w_{ij} = 0 \text{ if } j \notin \mathcal{N}_i. \quad (\star)$$

Define the (virtual) average $\bar{\mathbf{x}}^k = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i^k$, and if $\mathbf{W}$ is **column-stochastic** ($\sum_i w_{ij} = 1$), then average over i on both side of $(\star)$

$$\bar{\mathbf{x}}^{k+1} = \bar{\mathbf{x}}^k - \eta \frac{1}{n} \sum_{j=1}^n (\mathbf{A}_j \mathbf{x}_j^k - \mathbf{b}_j).$$



The average iterate follows a global gradient-like step on local gradients.

Quadratic objectives: gradient decomposition

We can decompose each local gradient around the average point:

$$\mathbf{A}_j \mathbf{x}_j^k - \mathbf{b}_j = \mathbf{A}_j \bar{\mathbf{x}}^k - \mathbf{b}_j + \mathbf{A}_j (\mathbf{x}_j^k - \bar{\mathbf{x}}^k).$$

Substituting into the update gives

$$\bar{\mathbf{x}}^{k+1} = \bar{\mathbf{x}}^k - \eta \left[\frac{1}{n} \sum_{j=1}^n (\mathbf{A}_j \bar{\mathbf{x}}^k - \mathbf{b}_j) \right] - \eta \frac{1}{n} \sum_{j=1}^n \mathbf{A}_j (\mathbf{x}_j^k - \bar{\mathbf{x}}^k).$$

Define the **consensus error term**:

$$\mathbf{E}_k := \frac{1}{n} \sum_{j=1}^n \mathbf{A}_j (\mathbf{x}_j^k - \bar{\mathbf{x}}^k).$$

Then
$$\bar{\mathbf{x}}^{k+1} = \bar{\mathbf{x}}^k - \eta \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\mathbf{x}}^k) - \eta \mathbf{E}_k.$$



Quadratic objectives: interpretation

Hence the average iterate evolves as

$$\bar{\mathbf{x}}^{k+1} = \bar{\mathbf{x}}^k - \eta \left(\frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\mathbf{x}}^k) + \mathbf{E}_k \right).$$

If the matrices $\mathbf{A}_i$ are uniformly bounded, say $\|\mathbf{A}_i\| \leq L$, then

$$\|\mathbf{E}_k\| \leq \frac{1}{n} \sum_{i=1}^n \|\mathbf{A}_i\| \|\mathbf{x}_i^k - \bar{\mathbf{x}}^k\| \leq L \max_i \|\mathbf{x}_i^k - \bar{\mathbf{x}}^k\|.$$

The virtual consensus point follows a gradient step on the *global objective* $\frac{1}{n} \sum_i f_i$, up to an error proportional to the consensus disagreement.

- Column-stochasticity ($\sum_i w_{ij} = 1$) preserves the average.
- The consensus error $\|\mathbf{E}_k\|$ decays with the spectral gap $1 - \lambda_2(W)$.
- Faster consensus $\Rightarrow$ smaller error $\Rightarrow$ near-centralized convergence.



Does DSGD converge? Insights from quadratic problems

- **Convergence of DSGD** depends on the consensus rate of the partial averaging and the convergence of PSGD.
- The consensus rate depends on the graph topology.
 - The fully connected graph reaches the consensus after one step.
 - For a ring graph, although not exact consensus, still closer.
- PSGD can tolerate inexact consensus if the error is decreasing.
- **Convergence guarantee? Speed?**



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Convergence of decentralized SGD



Partial averaging as one-step average consensus

- Write the DSGD in the matrix form.

DSGD

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^{(k)}; \xi_i^{(k)})$$

$$\mathbf{x}^{k+1} = \mathbf{W}\mathbf{x}^{k+\frac{1}{2}}$$

where $\mathbf{x}^l = [\mathbf{x}_1^l, \dots, \mathbf{x}_n^l]$, $l = k, k + \frac{1}{2}$.

- Recall average consensus algorithm (fixed weight matrix)

$$\mathbf{x}(t+1) = \mathbf{W}\mathbf{x}(t) = \mathbf{W}^t\mathbf{x}(0)$$

- The partial averaging in DSGD is one-step ($t = 1$) version of average consensus algorithm with varying $\mathbf{x}(0) = \mathbf{x}^{k+\frac{1}{2}}$



Review: Convergence rate of consensus

- Recall the average consensus algorithm (fixed weight matrix)

$$\mathbf{x}(t+1) = \mathbf{W}\mathbf{x}(t) = \mathbf{W}^t\mathbf{x}(0)$$

Theorem 1 (Convergence rate of average consensus)

If $\mathbf{W}$ is doubly stochastic, it holds for the average consensus protocol that

$$\left\| \mathbf{x}(t) - \frac{\mathbf{1}\mathbf{1}^T\mathbf{x}(t)}{n} \right\| \leq \rho^t \left\| \mathbf{x}(0) - \frac{\mathbf{1}\mathbf{1}^T\mathbf{x}(0)}{n} \right\|,$$

where $\rho = \max_{i \geq 2} |\lambda_i(\mathbf{W})| < 1$.

Q: Is one-step consensus enough for DSGD's convergence?



From average consensus to DSGD dynamics

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}), \quad \text{where } f_i(\mathbf{x}) = \mathbb{E}_{\xi_i \sim D_i} [F(\mathbf{x}; \xi_i)].$$

- Letting $\bar{\mathbf{x}}^{k+l} = \frac{\mathbf{1}\mathbf{1}^T \mathbf{x}^{k+l}}{n}$, $\ell = 1/2, 0$, partial averaging in DSGD:

$$\begin{aligned} \|\mathbf{x}^{k+1} - \bar{\mathbf{x}}^{k+1}\| &\leq \rho \left\| \mathbf{x}^{k+\frac{1}{2}} - \bar{\mathbf{x}}^{k+\frac{1}{2}} \right\| \quad (\text{because } \mathbf{x}^{k+1} = W\mathbf{x}^{k+\frac{1}{2}}) \\ &\leq \rho\eta \sum_{i=1}^n \left\| \nabla f_i(\mathbf{x}_i^k) - \frac{1}{n} \sum_{i=1}^n \nabla f_i(\mathbf{x}_i^k) \right\| + \rho\eta n\sigma \end{aligned}$$



From average consensus to DSGD dynamics

- Letting $\bar{\mathbf{x}}^{k+l} = \frac{\mathbf{1}\mathbf{1}^T \mathbf{x}^{k+l}}{n}$, $\ell = 1/2, 0$, partial averaging in DSGD:

$$\begin{aligned} & \|\mathbf{x}^{k+1} - \bar{\mathbf{x}}^{k+1}\| \\ & \leq \rho\eta \sum_{i=1}^n \left\| \nabla f_i(\mathbf{x}_i^k) - \frac{1}{n} \sum_{i=1}^n \nabla f_i(\mathbf{x}_i^k) \right\| + \rho\eta n\sigma \\ & \leq \rho\eta \sum_{i=1}^n \|\nabla f_i(\bar{\mathbf{x}}^k) - \nabla f_i(\mathbf{x}_i^k)\| + \rho\eta \sum_{i=1}^n \|\nabla f_i(\bar{\mathbf{x}}^k) - \nabla f(\bar{\mathbf{x}}^k)\| \\ & \quad + \rho\eta \sum_{i=1}^n \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\mathbf{x}_i^k) - \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\mathbf{x}}^k) \right\| + \rho\eta n\sigma \\ & \leq 2\rho\eta L \|\bar{\mathbf{x}}^k - \mathbf{x}^k\| + \rho\eta b + \rho\eta n\sigma \end{aligned}$$

where $b^2 \triangleq \max_{\mathbf{x}} \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}) - \nabla f(\mathbf{x})\|^2$ and L is the smoothness.



From average consensus to DSGD dynamics

- Letting $\bar{\mathbf{x}}^{k+l} = \frac{\mathbf{1}\mathbf{1}^T \mathbf{x}^{k+l}}{n}$, $\ell = 1/2, 0$, partial averaging in DSGD:

$$\|\mathbf{x}^k - \bar{\mathbf{x}}^k\| \leq (2\rho\eta L)^k \|\bar{\mathbf{x}}^0 - \mathbf{x}^0\| + \frac{\rho\eta(b + n\sigma)}{1 - 2\rho\eta L} \rightarrow 0 \quad \text{when } \eta \rightarrow 0$$

where $b^2 \triangleq \max_{\mathbf{x}} \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}) - \nabla f(\mathbf{x})\|^2$ and L is the smoothness.



Convergence rate of DSGD

Assumptions:

- f_i : L -smooth
- $\nabla F(\mathbf{x}; \xi_i)$ is an **unbiased estimate** of $\nabla f_i(\mathbf{x})$, with **bounded variance**
- Bounded data heterogeneity: $\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x) - \nabla f(x)\|^2 \leq b^2$
- The weight matrix **W** is doubly stochastic

Theorem 1 (Convergence rate of DSGD)

Suppose above assumptions hold and let $\eta = 1/\sqrt{K}$. Then

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\nabla f(\bar{\mathbf{x}}^k)\|_2^2] \leq \mathcal{O} \left(\frac{\sigma}{\sqrt{nK}} + \frac{\rho^{2/3} \sigma^{2/3}}{K^{2/3} (1 - \rho)^{1/3}} + \frac{\rho^{2/3} b^{2/3}}{K^{2/3} (1 - \rho)^{2/3}} \right)$$



Convergence of DSGD

Theorem 1 (Convergence rate of DSGD)

Suppose above assumptions hold and let $\eta = 1/\sqrt{K}$. Then

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\nabla f(\bar{\mathbf{x}}^k)\|_2^2] \leq \mathcal{O} \left(\frac{\sigma}{\sqrt{nK}} + \frac{\rho^{2/3} \sigma^{2/3}}{K^{2/3} (1 - \rho)^{1/3}} + \frac{\rho^{2/3} b^{2/3}}{K^{2/3} (1 - \rho)^{2/3}} \right)$$

- The first term dominates when $K \rightarrow \infty$, which suggests DSGD achieves *linear speedup* asymptotically.
- Convergence of DSGD depends on the *network topology*: sparse topology ($\rho \rightarrow 1$) results in slower convergence.
- Convergence of DSGD depends on the *data heterogeneity*: large heterogeneity b results in slower convergence.

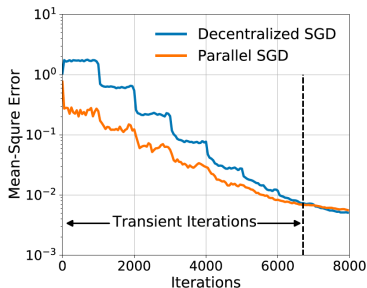


PSGD v.s. DSGD

PSGD: $\mathcal{O}\left(\frac{\sigma}{\sqrt{nK}}\right)$ **Transient iterations:** Extra overhead $\leq \frac{\sigma}{\sqrt{nK}}$

Extra overhead

DSGD: $\mathcal{O}\left(\frac{\sigma}{\sqrt{nK}} + \frac{\rho^{2/3}\sigma^{2/3}}{K^{2/3}(1-\rho)^{1/3}} + \frac{\rho^{2/3}b^{2/3}}{K^{2/3}(1-\rho)^{2/3}}\right)$



- DSGD can asymptotically converge as fast as P-SGD

- Transient iterations:

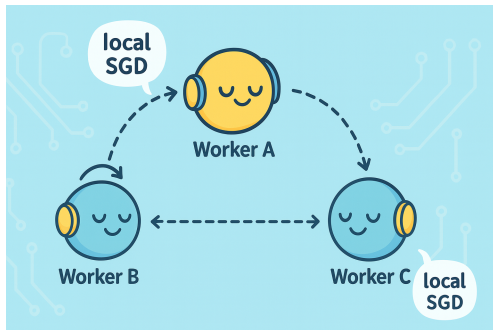
$$\mathcal{O}\left(\frac{\rho^4 n^3}{\sigma^2(1-\rho)^2} + \frac{\rho^4 n^3 b^4}{\sigma^6(1-\rho)^4}\right)$$

affected by *network topology* and *data heterogeneity*.



Summary of DSGD

- **Principle:** In each time step, nodes update its state by local SGD and communicate with its neighbors to perform partial averaging.
- **Benefit:** Low per-iter. communication cost when graph is sparse; achieve consensus asymptotically.
- **Drawback:** Cannot handle heterogeneous data setting.



Recap and fine-tuning

- What we have talked about **today**?
 - ⇒ **Decentralized SGD** reduces spatial communication reduction via partial averaging with neighbors.
 - ⇒ **Decentralized SGD** achieves the same convergence rate as Parallel SGD asymptotically.



Welcome anonymous survey!

